

MAHDI PANAHI

Helsingborg, Sweden · 070 449 42 63 · swemahdi@gmail.com · mahpan22@student.hh.se ·
github.com/Mahpan2222 · [My Portfolio](#)

■ PROFILE

Computer Engineering student at Halmstad University, currently completing a paid industry thesis at Autoliv with remaining coursework in the programme. Co-author of an IEEE conference paper on cryptographic performance for V2X vehicle safety systems, accepted at IEEE ETFA 2026. Comfortable in Linux and Windows environments, Python, and systematic testing and validation of embedded real-time systems. Over three years of part-time work alongside full-time studies.

■ PUBLICATIONS & RESEARCH

A Micro-benchmark of ECDSA Curves and Ed25519 for Low-Latency V2X Message Authentication

2025 – 2026

IEEE ETFA 2026 (Emerging Technologies & Factory Automation), Västerås, Sweden · Status: Accepted

- Co-authored with researchers at Halmstad University and Federal University of Rio Grande do Sul, Brazil
- Supervised by Edison Pignaton de Freitas (Halmstad University & UFRGS, Brazil), Wagner Ourique de Moraes and Alexandre dos Santos Roque
- Benchmarked three elliptic curve cryptographic algorithms (ECDSA secp256r1, brainpoolP256r1, Ed25519) for real-time V2X message authentication
- Key finding: Ed25519 achieves up to 27,000 signatures per second — fast enough for dense traffic within the strict 100 ms latency limit
- Implemented using Java Microbenchmark Harness (JMH) and the Bouncy Castle cryptography library
- Source code: github.com/Mahpan2222

■ EDUCATION

B.Sc. Computer Engineering — Datateknik

2023 – present

Halmstad University (Högskolan i Halmstad), Halmstad

- Programme: Dataingenjör inom datateknik — covering embedded systems, sensor fusion, algorithms, computer networks and software development
- Industry thesis in collaboration with Autoliv — see Degree Thesis below

■ DEGREE THESIS

Posture Estimation for Motorcycle Riders — Multi-IMU System

Feb 2025 – present

Autoliv, Sweden · Supervisor: Oscar Amador Molina

- Building a real-time posture estimation system using seven IMUs fitted into a motorcycle jacket and helmet
- Comparing Madgwick, Mahony and Extended Kalman Filter algorithms to handle sensor drift and keep orientation accurate over time
- Validating results against a video-based ground truth built with MediaPipe, measuring accuracy with RMSE of joint angles
- Responsible for the full hardware stack — ESP32 microcontrollers, microROS firmware and Wi-Fi data streaming
- Paid thesis position under a signed agreement with Autoliv

■ EXPERIENCE

Plant & Retail Associate

2023 – present

Blomsterlandet, Helsingborg · via Job and Talent

- Over three years of part-time work alongside full-time studies, at one of Sweden's largest garden and plant retail chains

- Day-to-day responsibilities include stock management, customer guidance and keeping the department in good order during busy periods
- Good working relationships across the team and with management — five managers have offered to act as references

Restaurant & Service Work

2019 – 2023

Multiple employers, Helsingborg area

- Worked summers from the age of 17 across various restaurant and service roles — learned to work under pressure and adapt quickly to new teams

Renovation & Painting Work (seasonal)

2020 – 2023

Private clients, Helsingborg area

- Took on interior painting and renovation jobs during summers, handling client contact and the work independently from start to finish

■ TECHNICAL SKILLS

Languages	Python · Java · C / C++ · PHP · JavaScript · Bash
Embedded	ESP32 · microROS · IMU integration
Networking	TCP/IP · Wi-Fi data streaming · V2X communication
Algorithms	Extended Kalman Filter · Madgwick · Mahony · Sensor fusion
Machine Learning	Supervised learning · Classification · scikit-learn
Frontend	HTML · CSS · JavaScript
Databases	SQL · MySQL · Database design
Tools	Linux · Windows · Git · ROS · MediaPipe
Methods	Testing & validation · Debugging · Agile teamwork · Technical documentation

■ LANGUAGES

Swedish	Fluent — professional working level
English	Fluent — professional working level
Farsi	Native

■ REFERENCES

Obey

Department Manager · Blomsterlandet

Josefin

Operations Manager · Blomsterlandet

Harald

Operations Manager · Blomsterlandet

Amina

Manager · Job and Talent

Christian Nils

Product Manager · Autoliv

Full contact details gladly provided upon request.

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